



CIE A Level Chemistry



Your notes

2.3 Formulas

Contents

- * Formulae
- * Balancing Equations
- * Empirical & Molecular Formulae
- * Water of Crystallisation



Your notes

Formulae

Formulae of Ionic Compounds

- **Ionic compounds** are formed from a **metal** and a **nonmetal** bonded together
- Ionic compounds are electrically neutral; the positive charges equal the negative charges

Charges on positive ions

- All metals form **positive** ions
 - There are some non-metal positive ions such as ammonium, NH_4^+ , and hydrogen, H^+
- The **metals** in Group 1, Group 2 and Group 13 have a charge of 1+ and 2+ and 3+ respectively
- The charge on the ions of the **transition elements can vary** which is why **Roman numerals** are often used to indicate their charge
- **Roman numerals** are used in some compounds formed from transition elements to show the **charge** (or **oxidation state**) of metal ions
 - Eg. in copper(II) oxide, the copper ion has a charge of 2+ whereas in copper(III) nitrate, the copper has a charge of 3+

Non-metal ions

- The **non-metals** in Group 15 to 17 have a negative charge and have the suffix '**ide**'
 - Eg. nitride, chloride, bromide, iodide
- Elements in Group 17 gain 1 electron so have a 1- charge, eg. Br^-
- Elements in Group 16 gain 2 electrons so have a 2- charge, eg. O^{2-}
- Elements in Group 15 gain 3 electrons so have a 3- charge, eg. N^{3-}
- There are also more **complex** negative ions, which are negative ions made up of more than one type of atom

Periodic table outlining common ion charges



Your notes

GROUP								18
1	2	H^+	13	14	15	16	17	NONE
Li^+	Be^{2+}					O^{2-}	F^-	NONE
Na^+	Mg^{2+}		Al^{3+}			S^{2-}	Cl^-	NONE
K^+	Ca^{2+}	TRANSITION ELEMENTS	Ga^{3+}				Br^-	NONE
Rb^+	Sr^{2+}						I^-	NONE

Copyright © Save My Exams. All Rights Reserved

The charges of simple ions depend on their position in the Periodic Table

Formulae of ionic compounds table

Ion	Formula and charge
Silver(I)	Ag^+
Ammonium	NH_4^+
Zinc(II)	Zn^{2+}
Hydroxide	OH^-
Nitrate	NO_3^-
Sulfate	SO_4^{2-}
Carbonate	CO_3^{2-}
Hydrogen carbonate	HCO_3^-

Phosphate

 PO_4^{3-} 

Your notes

 Worked example

Determine the formulae of the following ionic compounds:

- Magnesium chloride
- Iron(III) oxide
- Aluminium nitrate

Answer 1: Magnesium chloride

- Magnesium is in Group 2 so has a charge of 2+
- Chlorine is in Group 17 so has a charge of 1-
- Magnesium needs two chlorine atoms for each magnesium atom to be balanced so the formula is **MgCl₂**

Answer 2: Iron (III) oxide

- The Roman numeral states that iron has a charge of 3+
- Oxygen is in Group 16 so has a charge of 2-
- The charges need to be equal so 2 iron to 3 oxygen atoms will balance electrically, so the formula is **Fe₂O₃**

Answer 3: Aluminium nitrate

- Aluminium is in Group 13 so has a charge of 3+
- Nitrate is a **compound ion** and has a charge of 1-
- The complex ion needs to be placed in a bracket if more than 1 is needed
- The formula of aluminium nitrate is **Al(NO₃)₃**

 Exam Tip

Remember: **Compound ions** are ions that contain more than one type of element, such as OH⁻



Your notes

Balancing Equations

Balancing Equations

- A **symbol** equation is a shorthand way of describing a chemical reaction using **chemical symbols** to show the number and type of each atom in the reactants and products
- A **word** equation is a longer way of describing a chemical reaction using only **words** to show the reactants and products

Balancing equations

- During chemical reactions, atoms cannot be **created** or **destroyed**
- The number of each atom on each side of the reaction must therefore be the **same**
 - E.g. the reaction needs to be **balanced**
- When balancing equations remember:
 - Not to change any of the formulae
 - To put the numbers used to balance the equation **in front** of the formulae
 - To balance firstly the carbon, then the hydrogen and finally the oxygen in **combustion reactions** of organic compounds
- When balancing equations follow the following the steps:
 - Write the formulae of the reactants and products
 - Count the numbers of atoms in each reactant and product
 - Balance the atoms one at a time until all the atoms are balanced
 - Use appropriate state symbols in the equation
- The **physical state** of reactants and products in a chemical reaction is specified by using **state symbols**
 - **(s)** solid
 - **(l)** liquid
 - **(g)** gas
 - **(aq)** aqueous

Ionic equations

- In aqueous solutions, ionic compounds **dissociate** into their ions
- Many chemical reactions in aqueous solutions involve ionic compounds, however, only some of the ions in solution take part in the reactions
- The ions that do **not** take part in the reaction are called **spectator ions**
- An **ionic equation** shows **only** the ions or other particles taking part in a reaction, without showing the spectator ions



Your notes

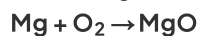
Worked example

Balance the following equation:

magnesium + oxygen $\rightarrow$ magnesium oxide

Answer:

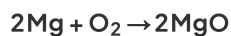
- **Step 1:** Write out the symbol equation showing reactants and products



- **Step 2:** Count the number of atoms in each reactant and product

	Mg	O
Reactants	1	2
Products	1	1

- **Step 3:** Balance the atoms one at a time until all the atoms are balanced



This is now showing that 2 moles of magnesium react with 1 mole of oxygen to form 2 moles of magnesium oxide

- **Step 4:** Use appropriate **state symbols** in the fully balanced equation

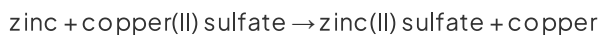




Your notes

Worked example

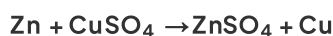
1. Balance the following equation:



2. Write the ionic equation for the above reaction.

Answer 1:

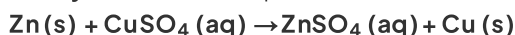
- Step 1:** To balance the equation, write out the symbol equation showing reactants and products



- Step 2:** Count the number of atoms in each reactant and product. The equation is already balanced

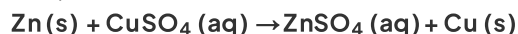
	Zn	Cu	S	O
Reactants	1	1	1	4
Products	1	1	1	4

- Step 3:** Use appropriate **state symbols** in the equation

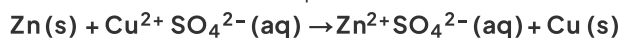


Answer 2:

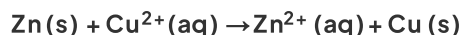
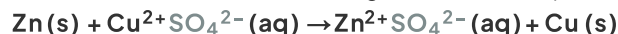
- Step 1:** The full chemical equation for the reaction is



- Step 2:** Break down reactants into their respective ions



- Step 3:** Cancel the spectator ions on both sides to give the ionic equation





Your notes

Empirical & Molecular Formulae

Empirical & Molecular Formulae

- The **molecular formula** is the formula that shows the **number** and **type** of each atom in a molecule
 - Eg. the molecular formula of ethanoic acid is $\text{C}_2\text{H}_4\text{O}_2$
- The **empirical formula** is the simplest whole number ratio of the elements present in one molecule or formula unit of the compound
 - Eg. the empirical formula of ethanoic acid is CH_2O
- **Organic molecules** often have **different** empirical and molecular formulae
- **Simple inorganic molecules** however have **often similar** empirical and molecular formulae
- **Ionic compounds always** have **similar** empirical and molecular formulae



Your notes

Empirical & Molecular Formulae Calculations

Empirical formula

- The **empirical formula** is the **simplest whole-number ratio** of the elements present in one molecule or formula unit of the compound
- It is calculated from knowledge of the ratio of masses of each element in the compound
- The empirical formula can be found by determining the **mass** of each element present in a sample of the compound
- It can also be deduced from data that gives the **percentage compositions by mass** of the elements in a compound

Worked example

Calculating empirical formula from mass

Determine the empirical formula of a compound that contains 10 g of hydrogen and 80 g of oxygen.

Answer:

Elements	Hydrogen	Oxygen
Mass of each element (g)	10	80
Atomic mass	1.0	16.0
Moles = mass / A_r	$\frac{10}{1.0} = 10$	$\frac{80}{16.0} = 5$
Ratio (divide by smallest value)	$\frac{10}{5} = 2$	$\frac{5}{5} = 1$

- So, the empirical formula of the compound is H_2O



Your notes

Worked example

Calculating empirical formula from %

Determine the empirical formula of a compound that contains 85.7% carbon and 14.3% hydrogen.

Answer:

Elements	Carbon	Hydrogen
Mass of each element (%)	85.7	14.3
Atomic mass	12.0	1.0
Moles = mass / A_r	$\frac{85.7}{12.0} = 7.14$	$\frac{14.3}{1.0} = 14.3$
Ratio (divide by smallest value)	$\frac{7.14}{7.14} = 1$	$\frac{14.3}{7.14} = 2.00$

- So, the empirical formula of the compound is CH_2

Molecular formula

- The **molecular formula** gives the exact number of atoms of each element present in the formula of the compound
- The molecular formula can be found by dividing the **relative formula mass** of the **molecular formula** by the **relative formula mass** of the **empirical formula**
- Multiply** the number of each element present in the empirical formula by this number to find the molecular formula



Your notes

 Worked example**Calculating molecular formula**

The empirical formula of **X** is $\text{C}_4\text{H}_{10}\text{S}$ and the relative formula mass of **X** is 180.

What is the molecular formula of **X**?

Relative atomic mass Carbon:12 Hydrogen:1 Sulfur:32

Answer:

- **Step 1:** Calculate the relative formula mass of the empirical formula
 - Relative formula mass = $(\text{C} \times 4) + (\text{H} \times 10) + (\text{S} \times 1)$
 - Relative formula mass = $(12 \times 4) + (1 \times 10) + (32 \times 1)$
 - Relative formula mass = 90
- **Step 2:** Divide the relative formula mass of **X** by the relative formula mass of the empirical formula
 - Ratio between M_r of **X** and the M_r of the empirical formula = $180/90$
 - Ratio between M_r of **X** and the M_r of the empirical formula = 2
- **Step 3:** Multiply each number of elements by 2
 - $(\text{C}_{4 \times 2}) + (\text{H}_{10 \times 2}) + (\text{S}_{1 \times 2}) = (\text{C}_8) + (\text{H}_{20}) + (\text{S}_2)$
 - The molecular formula of **X** is $\text{C}_8\text{H}_{20}\text{S}_2$



Your notes

Water of Crystallisation

Water of Crystallisation

- **Water of crystallisation** is when some compounds can form **crystals** which have **water** as part of their structure
- A compound that contains water of crystallisation is called a **hydrated compound**
- The water of crystallisation is separated from the main formula by a **dot** when writing the chemical formula of hydrated compounds
 - E.g. hydrated copper(II) sulfate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- A compound which doesn't contain water of crystallisation is called an **anhydrous compound**
 - E.g. anhydrous copper(II) sulfate is CuSO_4
- A compound can be hydrated to **different degrees**
 - E.g. cobalt(II) chloride can be hydrated by **six** or **two** water molecules
 - $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ or $\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$
- The conversion of anhydrous compounds to hydrated compounds is reversible by heating the hydrated salt:
$$\text{Hydrated: } \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightleftharpoons \text{CuSO}_4 + 5\text{H}_2\text{O} \quad \text{:Anhydrous}$$
- The degree of hydration can be calculated from experimental results:
 - The mass of the hydrated salt must be measured before heating
 - The salt is then heated until it reaches a constant mass
 - The two mass values can be used to calculate the number of moles of water in the hydrated salt - known as the water of crystallisation



Your notes

Worked example

Calculating water of crystallisation

10.0 g of hydrated copper sulfate are heated to a constant mass of 5.59 g.

Determine the formula of the original hydrated copper sulfate, $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$.

(M_r data: $\text{CuSO}_4 = 159.6$, $\text{H}_2\text{O} = 18.0$)

Answer:

Compounds	CuSO_4	H_2O
Mass of each compound (g)	5.59	4.41
Formula mass	159.6	18.0
Moles = mass / M_r	$\frac{5.59}{159.6} = 0.035$	$\frac{4.41}{18.0} = 0.245$
Ratio (divide by smallest value)	$\frac{0.035}{0.035} = 1$	$\frac{0.245}{0.035} = 7$

- So, the value of x is 7
- Therefore, the formula of the hydrated salt is $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$

Exam Tip

A water of crystallisation calculation can be completed in a similar fashion to an empirical formula calculation

- Instead of elements, you start with the salt and water
- Instead of dividing by atomic masses, you divide by molecular / formula masses
- The rest of the calculation works the same way as the empirical formula calculation